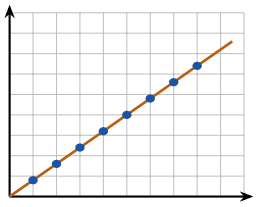
Name: _____

Class: _____

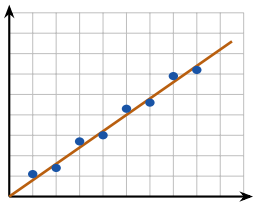
Date: _____ **Mr. Rawson • GCSE Physics**

Describing a Graph: Correlation

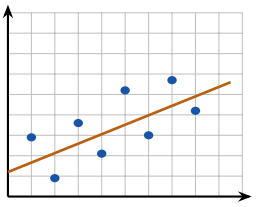
Correlation describes how closely the points follow the trendline. Look at the points, then pick the word from the row that matches what you see.



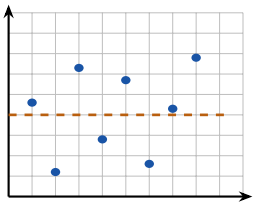
Perfect
Every point sits **on** the line.



Strong
Every point sits **close to** the line.



Weak
The points follow the line, but sit **well away** from it.

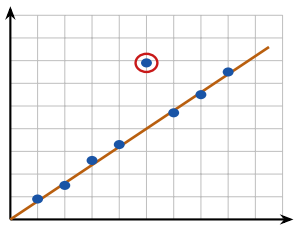


None
The points make **no** pattern.

Sentences to copy and complete

Correlation	Sample sentence
Perfect	The correlation is **perfect** because every point sits **exactly on** the trendline.
Strong	The correlation is **strong** because every point sits **close to** the trendline.
Weak	The correlation is **weak** because the points follow the trendline, but several sit **well away from** it.
None	There is **no correlation** because the points are **spread out** and make no pattern, so there is no trendline to follow.

An anomaly



Anomaly is a point that sits clearly away from the pattern that the other points make. Circle it, and give its run number.

One anomaly: **Run 5** is an anomaly because it sits clearly away from the trendline. The other points sit close to it.

No anomaly: There are no anomalies, because every point sits close to the pattern the points make.

Name: _____ **Date:** _____

Linear and inverse relationships: three parts, three sentences

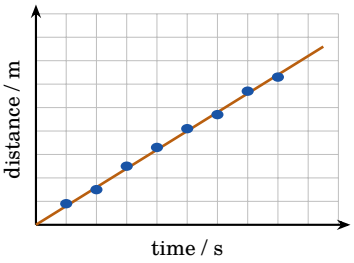
Every description of a graph has the same three parts. Colour each part of your own writing the same way.

1. The relationship
direct (linear) or inverse

2. The correlation
perfect, strong, weak or none, and why

3. The anomalies
how many, and what the other points do

A toy car: a straight line through the origin — as time doubles, distance doubles.

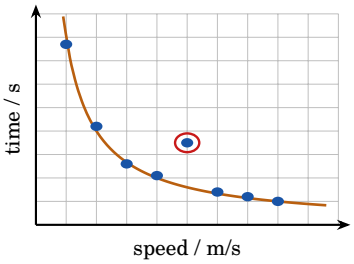


1 The graph shows a **direct (linear)** relationship: as time **increases**, distance **increases**.

2 The correlation is **strong**, because every point sits close to the trend-line.

3 There are **no anomalies**, because every point follows the trend.

A journey of fixed length: a curve that falls and flattens out — as speed doubles, time halves.

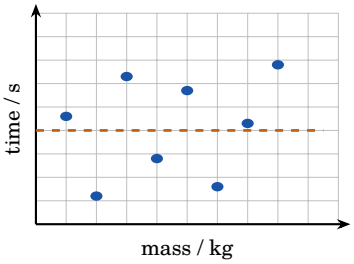


1 The graph shows an **inverse** relationship: as speed **increases**, time **decreases**.

2 The correlation is **strong**, because the points sit close to the curve of the trendline.

3 There is **one anomaly**, run 5, which sits clearly away from the curve, but the **other points fall in line with the expected trend**.

Points with no pattern — the graph does not show a relationship.



1 The graph shows **no clear relationship**: as mass increases, time goes up and down with no pattern.

2 There is **no correlation**, because the points are spread out and no line fits them.

3 I cannot name an anomaly, because **no pattern** exists for a point to sit away from. The **whole data set** is scattered.

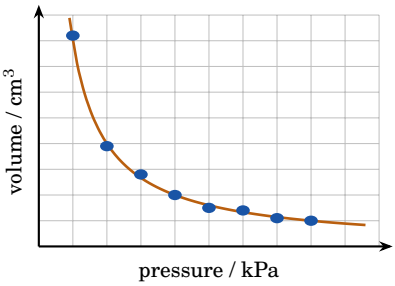
Join the three parts. Put the sentences together with a short link: *The graph shows an inverse relationship, and the correlation is strong. There is one anomaly, run 5, but the other points fall in line with the expected trend.* The word *and* adds a second fact, and the word *but* marks the anomaly against the trend.

Name: _____ **Date:** _____

Choose your words

Part	Pick one from each group of words
1. Relationship	The graph shows a { **direct (linear)** | **inverse** | **no clear** } relationship: as _____ { **increases** | **decreases** }, _____ { **increases** | **decreases** }.
2. Correlation	The correlation is { **perfect** | **strong** | **weak** }, because { **every point sits on** | **every point sits close to** | **the points sit well away from** } the trendline. There is **no correlation**, because the points are spread out with no pattern.
3. Anomalies	There are **no anomalies**, because every point follows the trend. There is **one anomaly**, run ____, which sits clearly away from the { **line** | **curve** }, but the **other points fall in line with the expected trend**. There are **several anomalies**, runs _____, so the pattern is **not clear**.

Your turn



Describe this graph in three sentences, one for each part. Use the colours: blue, amber, green.

1. Relationship.

The graph shows an **inverse** relationship: as pressure **increases**, volume **decreases**.

2. Correlation.

The correlation is **strong**, because every point sits close to the curve of the trendline.

3. Anomalies.

There are **no anomalies**, because every point follows the trend.

Now join them into two sentences, with *and* and *but*.

The graph shows an inverse relationship, and the correlation is strong. There are no anomalies, because every point follows the trend.